

Joseph Larry Austerweil

Curriculum Vitae

September 2026

📍 School of Design & Science, Chiba Institute of Technology,
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Education

2007	Sc.B (Hons)	Applied Mathematics-Computer Science	Brown University
2011	MA	Statistics	University of California, Berkeley
2012	PhD	Psychology	University of California, Berkeley

Positions held

2026–	Professor and Academic Director (co-founding faculty member), School of Design & Science, Chiba Tech
2025–	Principal Investigator , Computational Mind Laboratory, Henkaku Center, Chiba Tech (https://cml.chibatech.dev)
2025–2026	Senior Research Scientist , Center for Innovation, Chiba Tech
2023–2025	Associate Professor (affiliate), Department of Radiology, University of Wisconsin – Madison
2022–	Visiting Researcher , RIKEN Advanced Intelligence Project, Tokyo, Japan.
2021–2025	Associate Professor , Department of Psychology, University of Wisconsin – Madison.
2022–2025	Associate Professor (affiliate), Department of Computer Science, University of Wisconsin – Madison.
2019–	Associate Editor , <i>Computational Brain & Behavior</i>
2016–2021	Assistant Professor , Department of Psychology, University of Wisconsin – Madison.
2016–2021	Assistant Professor (affiliate), Department of Computer Science, University of Wisconsin – Madison.
2013–2016	Assistant Professor , Department of Cognition, Linguistics, & Psychological Sciences, Brown University.
2013	Postdoctoral Researcher with Noah Goodman, Stanford University.

Recent honors and awards

2019	William K Estes Early Career Award
2019	L & S Distinguished Honors Faculty Award
2019	IBM Watson AI xPrize Final Top 32 Team (Co-I of Team Brown HCRI)
2018	Vilas Life Cycle Professorship
2018	University Housing Honored Instructor
2017	Fellow, Psychonomic Society

Open Science

My laboratory posts all of its code and research data on its GitHub group website (<https://github.com/AusterweilLab/>) and/or OSF repositories.

Grants

Near decade-long track record of funding from internal, non-profit, industry, private philanthropic, and different federal agencies totaling over \$1 million dollars in total costs (sub-PI, co-PI, and PI only).

Current Grants

Private Donor (gift to Chiba Institute of Technology)

Development of AI Education Content

Co-I: Austerweil (\$10,000,000)

2026–2029

Corporate Sponsor (gift to Chiba Institute of Technology)

Supporting the Evaluation of AI

Co-I: Austerweil (\$100,000)

2026

Japanese Probabilistic Computing Consortium Association (JPCCA)

A Narrative Introduction to Probability (open, bilingual textbook)

PI: Austerweil

2025–

Grants Under Review

JST-ANR CREST, FY2026 (Symbiotic AI Interdisciplinary System; France–Japan joint call)

AI-Mediated Consensus Decision-Making in Human Groups and Networks

Co-PI: Austerweil

10/2026–03/2032

JSPS KAKENHI, Grant-in-Aid for Scientific Research (B), FY2027 competition
 Computational Principles of Long-Term Learning and Transfer through Human-AI Co-Adaptation
 PI: Austerweil FY2027–FY2030

JSPS KAKENHI, Grant-in-Aid for Challenging Research (Exploratory), FY2027 competition
 Quantitative Laws of Trust Calibration in AI: The Psychophysics of Fluency and Recalibration in Artificial Agents
 PI: Austerweil FY2027–FY2029

Completed Grants

NIH NIA
 PET/MR Correlates of Accelerated Aging in Chronic Epilepsy
 PI: Mcmillan/Prabhakaran – Co-I: Austerweil (\$3,044,075) 04/15/21–03/31/26

Army Research Office
 Human-Guided ML for Futuristic Human-Machine Teaming
 Sub-PI: Austerweil (\$363,450) 02/01/22–01/31/25

Templeton Foundation (through Duke University)
 Rumination: Necessary or disordered computation?
 PI: Austerweil (\$23,000) 07/31/23–03/31/24

NIH NICHD
 Prediction and Early Language Development in Young Children with ASD
 PI: Weismer/Saffran – Co-I: Austerweil (\$3,496,805) 07/01/19–06/30/24

American Family Insurance
 Continuation of Question Asking with Differing Knowledge and Goals
 PI: Austerweil (\$60,356) 01/01/21–08/31/21

American Family Insurance
 Question Asking with Differing Knowledge and Goals
 PI: Austerweil (\$75,000) 01/01/20–12/31/20

Wisconsin Alumni Research Foundation
 Fall Competition Award
 PI: Austerweil (\$46,655) 07/01/19–06/30/20

DARPA
 Homo SocioNeticus: Scaling the cognitive foundations of online behavior
 PI: Orr (\$5,600,000) – Sub-PI: Austerweil (\$218,782) 10/30/17–06/30/19

NIH NIA R21
 Computational modeling of semantic decline in Alzheimer's disease
 PI: Austerweil/Zemla (\$400,000) 09/01/16–05/31/19

DARPA
 Foundations of Human-Machine Collaboration: Networks of Social and Moral Norms in Human and Artificial Agents
 PI: Malle co-PI: Austerweil (\$413,092) 04/15/16–01/14/17

Rhode Island Foundation
 Examining How Autism Spectrum Disorders Affect Social Causal Learning
 PI: Austerweil (\$25,000) 04/01/16–06/30/16

Publication Highlights

- * Over 70 peer-reviewed papers that have been published in journals, prestigious conferences, or books.
- * A current advisee (Zhuolun Zhong) led all three of the laboratory's papers at the 2026 Annual Meeting of the *Cognitive Science Society*, one of which was accepted for oral presentation (top ~10% of submissions).

- * Two published journal articles in 2020 with UW-Madison undergraduate research assistants as co-authors, one of which is an extended version of their honors thesis in my laboratory.
- * A former PhD advisee (Mark Ho) received a talk at the extremely prestigious at *Neural Information Processing Systems (NeurIPS)* for a peer-reviewed paper on his work in my laboratory. Acceptance with oral presentation is <2% of all submissions.

Publications

Published Peer-Reviewed Journal Articles

1. Austerweil, J. L., Liew, S. X., Conaway, N., & Kurtz, K. J. (2024). Creating something different: Similarity, contrast, and representativeness in categorization. *Computational Brain & Behavior*, 1–35.
2. Zemla, J. C., Gooding, D. C., & Austerweil, J. L. (2023). Evidence for optimal semantic search throughout adulthood. *Nature: Scientific Communications*, **13**, 22528.
3. Ho, M. K., Cushman, F., Littman, M. L., & Austerweil, J. L. (2021). Communication in action: Planning and interpreting communicative demonstrations. *Journal of Experimental Psychology: General*, **150**(11), 2246–2272.
4. Mohanta, S., Afrasiabi, M., Casey, C., Tanabe, S., Redinbaugh, M. J., Kambi, N. A., Phillips, J. M., Polyakov, D., Filbey, W., Austerweil, J. L., Sanders, R. D., & Saalman, Y. B. (2021). Predictive feedback, early sensory representations, and fast responses to predicted stimuli depend on nmda receptors. *Journal of Neuroscience*, **41**(49), 10130–10147.
5. Sanborn, A. N., Heller, K., Austerweil, J. L., & Chater, N. (2021). REFRESH: A new approach to modelling dimensional biases in perceptual similarity and categorization. *Psychological Review*, **128**(6), 1145–1186.
6. Lange, K. V., Hopman, E. W. M., Zemla, J. C., & Austerweil, J. L. (2020). Evidence against a relation between bilingualism and creativity. *PLOS ONE*, **15**(6), 1–18.
7. Zemla, J. C., Cao, K., Mueller, K. D., & Austerweil, J. L. (2020). SNAFU: The Semantic Network and Fluency Utility. *Behavior Research Methods*, 1–19.
8. Austerweil, J. L., Sanborn, S. A., & Griffiths, T. L. (2019). Learning how to generalize. *Cognitive Science*, **43**(8), e12777.
9. Ho, M. K., Cushman, F., Littman, M. L., & Austerweil, J. L. (2019). People teach with rewards and punishments as communication not reinforcements. *Journal of Experimental Psychology: General*, **148**(3), 520–549.
10. Wulff, D. U., De Deyne, S., Jones, M. N., Mata, R., Austerweil, J. L., Baayen, R. H., Balota, Baronchelli, A., Brysbaert, M., Cai, Q., Dennis, S., Hills, T. T., Kenett, Y. N., Keuleers, E., Marelli, M., Pakhomov, S. V., Ramscar, M., Schooler, L. J., Shing, Y. L., ... Verissimo, J. (2019). New perspectives on the aging lexicon. *Trends in Cognitive Sciences*, **23**(8), 686–698.
11. Zemla, J. C., & Austerweil, J. L. (2019). Analyzing knowledge retrieval impairments associated with Alzheimer's disease using network analyses. *Complexity*, e4203158.
12. Griffiths, T. L., Daniels, D. D., Austerweil, J. L., & Tenenbaum, J. B. (2018). Subjective randomness as statistical inference. *Cognitive Psychology*, **103**, 85–109.
13. Zemla, J. C., & Austerweil, J. L. (2018). Estimating semantic networks of groups and individuals from fluency data. *Computational Brain & Behavior*, **1**(1), 36–58.
14. Austerweil, J. L., Palmer, S. E., & Griffiths, T. L. (2017). Learning to be (in)variant using prior knowledge: The case of orientation in shape recognition. *Cognitive Science*, **41**, 1183–1201.
15. Cibelli, E., Xu, Y., Austerweil, J. L., Griffiths, T. L., & Regier, T. (2016). The Sapir-Whorf hypothesis and probabilistic inference: Evidence from the domain of color. *PLOS: ONE*, **11**(8), e0161521.
16. Sobel, D. M., & Austerweil, J. L. (2016). Coding choices affect the analyses of a false belief measure. *Cognitive Development*, **40**, 9–23.
17. Abbott, J. T., Austerweil, J. L., & Griffiths, T. L. (2015). Random walks on semantic networks can resemble optimal foraging. *Psychological Review*, **122**(3), 558–569.
18. Austerweil, J. L. (2015). Contradictory "heuristic" theories of autism spectrum disorders: The case for theoretical precision using computational models. *Autism*, **19**(3), 367–368.
19. Cohen-Priva, U., & Austerweil, J. L. (2015). Analyzing the history of Cognition using topic models. *Cognition*, **135**, 4–9.
20. Austerweil, J. L., & Griffiths, T. L. (2013). Constructing flexible feature representations using nonparametric Bayesian inference. *Psychological Review*, **120**(4), 817–851.
21. Griffiths, T. L., & Austerweil, J. L. (2012). Bayesian generalization with circular consequential regions. *Journal of Mathematical Psychology*, **56**(4), 281–285.
22. Austerweil, J. L., & Griffiths, T. L. (2011). A rational model of the effects of distributional information on feature learning. *Cognitive Psychology*, **63**, 173–209.
23. Austerweil, J. L., & Griffiths, T. L. (2011). Seeking confirmation is rational for deterministic hypotheses. *Cognitive Science*, **35**, 499–526.

24. Gardner, J. S., Austerweil, J. L., & Palmer, S. E. (2010). Vertical position as a cue to pictorial depth: Height in the picture plane versus distance to the horizon. *Attention, Perception, & Psychophysics*, **72**, 445–453.

Peer-Reviewed Conference Proceedings

1. Austerweil, J. L., Nishikawa, J., & Morita, J. (2026). Rule-ambiguity and distributional play in asynchronous L2 shiritori. *Proceedings of the 43rd Annual Meeting of the Japanese Cognitive Science Society (International Session)*.
2. Hao, H., & Austerweil, J. L. (2026). TEMPO: Transformers for temporal disease progression from cross-sectional data. *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*, **333**, 1–31.
3. Malkoc, O., Hao, H., Szep, G., Saha, S., Oka, M., & Austerweil, J. L. (2026). Adaptive Bayesian active querying with LLMs for efficient information gathering. *Deploying AI in Healthcare (DAIH) Workshop at the Conference on Language Modeling (COLM)*.
4. Zhong, Z., Caspar, L., & Austerweil, J. L. (2026). How the teaching style and interpretation type of state interventions shape multi-agent coordination. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.
5. Zhong, Z., Caspar, L., & Austerweil, J. L. (2026). Individual differences in human teaching of reinforcement learning agents: Evidence from Bayesian hypothesis testing. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.
6. Zhong, Z., Prystawski, B., Wu, S. A., Fascendini, B., Saeedpour, S., & Austerweil, J. L. (2026). Interpretational alignment: How agents learn from physical guidance depends on how they interpret it. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.
7. Hao, H., & Austerweil, J. L. (2025). Bayesian event-based model for disease subtype and stage inference. *Proceedings of the Machine Learning for Health Conference 2025 (Vol. 297)*.
8. Hao, H., & Austerweil, J. L. (2025). Joint progression modeling for mixed pathologies (JPM): A probabilistic framework for mixed-pathology progression. *Proceedings of the Machine Learning for Health Conference 2025*.
9. Hao, H., Prabhakaran, V., Nair, V. A., Adluru, N., & Austerweil, J. L. (2025). Stage-aware event-based modeling (SA-EBM) for disease progression. In M. Agrawal, K. Deshpande, M. Engelhard, S. Joshi, S. Tang, & I. Urteaga (Eds.), *Proceedings of the 10th machine learning for healthcare conference* (Vol. 298). PMLR.
10. Malle, B. F., Austerweil, J. L., Chi, V. B., Kenett, Y. N., Beck, E. D., Thapa, S., & Allaham, M. (2021). Cognitive properties of norm representations. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society*, 819–825.
11. Y.-S. Chuang, Zhang, X., Ma, Y., Ho, M. K., Austerweil, J. L., & Zhu, X. (2021). Using machine teaching to investigate human assumptions when teaching reinforcement learners. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society*, 1159–1165.
12. Y.-S. Chuang, Hubbard, E., & Austerweil, J. L. (2020). The "fraction sense" emerges from a deep convolutional neural network. In S. Denison, M. Mack, Y. Yu, & B. C. Strong (Eds.), *Proceedings of the 42nd annual meeting of the cognitive science society* (pp. 1207–1213). Cognitive Science Society.
13. Afrasiabi, M., Orr, M. G., & Austerweil, J. L. (2019). Evaluating theories of collaborative cognition using the hawkes process and a large naturalistic data set. In A. Goel, C. Seifert, & C. Freska (Eds.), *Proceedings of the 41st annual meeting of the cognitive science society* (pp. 69–75). Cognitive Science Society.
14. Liew, S. X., & Austerweil, J. L. (2019). Novel categories are distinct from "not"-categories. In A. Goel, C. Seifert, & C. Freska (Eds.), *Proceedings of the 41st annual meeting of the cognitive science society* (pp. 2147–2153). Cognitive Science Society.
15. Payton, M., Zemla, J. C., & Austerweil, J. L. (2019). Subjective randomness in a non-cooperative game. In A. Goel, C. Seifert, & C. Freska (Eds.), *Proceedings of the 41st annual meeting of the cognitive science society* (pp. 2544–2549). Cognitive Science Society.
16. Cochrane, A., Austerweil, J. L., Simmering, V., & Green, C. S. (2018). Rapid learning in early attentional processing: Bayesian estimation of trial-by-trial updating. In T. T. Rogers, M. Rau, X. Zhu, & C. Kalish (Eds.), *Proceedings of the 40th annual meeting of the cognitive science society* (pp. 232–237). Cognitive Science Society.
17. Fathan, M. H., Renfro, E. K., Austerweil, J. L., & Beckage, N. M. (2018). Do humans navigate via random walks? modeling navigation in a semantic word game. In T. T. Rogers, M. Rau, X. Zhu, & C. Kalish (Eds.), *Proceedings of the 40th annual meeting of the cognitive science society* (pp. 366–371). Cognitive Science Society.
18. Ho, M. K., Littman, M. L., Cushman, F., & Austerweil, J. L. (2018). Effectively learning from pedagogical demonstrations. In T. T. Rogers, M. Rau, X. Zhu, & C. Kalish (Eds.), *Proceedings of the 40th annual meeting of the cognitive science society* (pp. 505–510). Cognitive Science Society.
19. Hopman, E. W. M., Thompson, B., Austerweil, J. L., & Lupyan, G. (2018). Predictors of l2 word learning accuracy: A big data investigation. In T. T. Rogers, M. Rau, X. Zhu, & C. Kalish (Eds.), *Proceedings of the 40th annual meeting of the cognitive science society* (pp. 515–518). Cognitive Science Society.

20. Conway, N., & Austerweil, J. L. (2017). PACKER: An exemplar model of category generation. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th annual meeting of the cognitive science society* (pp. 1812–1817). Cognitive Science Society.
21. Ho, M. K., Littman, M. L., & Austerweil, J. L. (2017). Teaching by intervention: Working backwards, undoing mistakes, or correcting mistakes? In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th annual meeting of the cognitive science society* (pp. 526–531). Cognitive Science Society.
22. Ren, J., & Austerweil, J. L. (2017). Interpreting asymmetries in speech perception in Bayesian inference. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th annual meeting of the cognitive science society* (pp. 1010–1015). Cognitive Science Society.
23. Sarathy, V., Scheutz, M., Kenett, Y., Allaham, M. M., Austerweil, J. L., & Malle, B. F. (2017). Mental representations and computational modeling of context-specific human norm systems. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th annual meeting of the cognitive science society* (pp. 1035–1040). Cognitive Science Society.
24. Zemla, J. C., & Austerweil, J. L. (2017). Modeling semantic fluency data as search on a semantic network. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th annual meeting of the cognitive science society* (pp. 3646–3651). Cognitive Science Society.
25. Austerweil, J. L., Brawner, S., Greenwald, A., Hilliard, E., Ho, M. K., Littman, M. L., MacGlashan, J., & Trimbach, C. (2016). The impact of outcome preferences in a collection of non-zero-sum grid games. *AAAI Spring System 2016 on Challenges and Opportunities in Multiagent Learning for the Real World*.
26. Ho, M. K., Littman, M. L., MacGlashan, J., Cushman, F., & Austerweil, J. L. (2016). Showing versus doing: Teaching by demonstration. *Advances in Neural Information Processing Systems* 29, 3027–3035.
27. Ho, M. K., MacGlashan, J., Greenwald, A., Littman, M., Hilliard, E., Trimbach, C., Brawner, S., Tenenbaum, J. B., Kleiman-Weiner, M., & Austerweil, J. L. (2016). Feature-based joint planning and norm learning in collaborative games. In A. Papafragou, D. Grodner, D. Mirman, & J. Trueswell (Eds.), *Proceedings of the 38th annual meeting of the cognitive science society* (pp. 1158–1163). Cognitive Science Society.
28. Kenett, Y. N., & Austerweil, J. L. (2016). Examining search processes in low and high creative individuals with random walks. In A. Papafragou, D. Grodner, D. Mirman, & J. Trueswell (Eds.), *Proceedings of the 38th annual meeting of the cognitive science society* (pp. 313–318). Cognitive Science Society.
29. Kleiman-Weiner, M., Ho, M. K., Austerweil, J. L., Littman, M. L., & Tenenbaum, J. B. (2016). Coordinate to cooperate or compete: Abstract goals and joint intentions in social interaction. In A. Papafragou, D. Grodner, D. Mirman, & J. Trueswell (Eds.), *Proceedings of the 38th annual meeting of the cognitive science society* (pp. 1679–1684). Cognitive Science Society.
30. Zemla, J. C., Kenett, Y., Jun, K.-S., & Austerweil, J. L. (2016). U-INVITE: Estimating individual semantic networks from fluency data. In A. Papafragou, D. Grodner, D. Mirman, & J. Trueswell (Eds.), *Proceedings of the 38th annual meeting of the cognitive science society* (pp. 1907–1912). Cognitive Science Society.
31. Ho, M. K., Littman, M. L., Cushman, F., & Austerweil, J. L. (2015). Teaching with rewards and punishments: Reinforcement or communication? In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, T. Matlock, C. D. Jennings, & P. P. Maglio (Eds.), *Proceedings of the 37th annual meeting of the cognitive science society* (pp. 920–925). Cognitive Science Society.
32. Malle, B. F., Scheutz, M., & Austerweil, J. L. (2015). Networks of social and moral norms in human and artificial agents. *International Conference on Robot Ethics*.
33. Qian, T., & Austerweil, J. L. (2015). Learning additive and substitutive features. In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, T. Matlock, C. D. Jennings, & P. P. Maglio (Eds.), *Proceedings of the 37th annual meeting of the cognitive science society* (pp. 1919–1924). Cognitive Science Society.
34. Austerweil, J. L. (2014). Testing the psychological validity of cluster construction biases. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.), *Proceedings of the 36th annual meeting of the cognitive science society* (pp. 122–127). Cognitive Science Society.
35. Jia, Y., Abbott, J. T., Austerweil, J. L., Griffiths, T. L., & Darrell, T. (2013). Visual concept learning: Combining machine vision and Bayesian generalization on concept hierarchies. In C. J. C. Burges, L. Bottou, M. Welling, Z. Ghahramani, & K. Q. Weinberger (Eds.), *Advances in neural information processing systems* 26 (pp. 1842–1850). Curran Associates, Inc.
36. Abbott, J. T., Austerweil, J. L., & Griffiths, T. L. (2012). Constructing a hypothesis space from the Web for large-scale Bayesian word learning. In N. Miyake, D. Peebles, & R. P. Cooper (Eds.), *Proceedings of the 34th annual meeting of the cognitive science society* (pp. 54–59). Cognitive Science Society.

37. Abbott, J. T., Austerweil, J. L., & Griffiths, T. L. (2012). Human memory search as a random walk in a semantic network. In F. Pereira, C. J. C. Burges, L. Bottou, & K. Q. Weinberger (Eds.), *Advances in neural information processing systems* 25 (pp. 3041–3049). Curran Associates, Inc.
38. Griffiths, T. L., Austerweil, J. L., & Berthiaume, V. G. (2012). Comparing the inductive biases of simple neural networks and Bayesian models. In N. Miyake, D. Peebles, & R. P. Cooper (Eds.), *Proceedings of the 34th annual meeting of the cognitive science society* (pp. 402–407). Cognitive Science Society.
39. Austerweil, J. L., Friesen, A. L., & Griffiths, T. L. (2011). An ideal observer model for identifying the reference frame of objects. In F. Pereira, C. J. C. Burges, L. Bottou, & K. Q. Weinberger (Eds.), *Advances in neural information processing systems* 24 (pp. 3041–3049). Curran Associates, Inc.
40. Austerweil, J. L., & Griffiths, T. L. (2010). An ideal observer model for identifying the reference frame of objects. In J. Lafferty, C. K. I. Williams, J. Shawne-Taylor, R. Zemel, & A. Culotta (Eds.), *Advances in neural information processing systems* 23 (pp. 82–90). Curran Associates, Inc.
41. Austerweil, J. L., & Griffiths, T. L. (2010). Learning hypothesis spaces and dimensions through concept learning. In S. Ohlsson & R. Catrambone (Eds.), *Proceedings of the 32nd annual meeting of the cognitive science society* (pp. 73–78). Cognitive Science Society.
42. Austerweil, J. L., & Griffiths, T. L. (2009). Analyzing human feature learning as nonparametric Bayesian inference. In D. Koller, Y. Bengio, D. Schuurmans, & L. Bottou (Eds.), *Advances in neural information processing systems* 21 (pp. 97–104). Curran Associates, Inc.
43. Austerweil, J. L., & Griffiths, T. L. (2009). The effect of distributional information on feature learning. In N. Taatgen & H. van Rijn (Eds.), *Proceedings of the 31st annual meeting of the cognitive science society* (pp. 2765–2770). Cognitive Science Society.
44. Austerweil, J. L., & Griffiths, T. L. (2008). A rational analysis of confirmation with deterministic hypotheses. In B. C. Love, K. McRae, & V. M. Sloutsky (Eds.), *Proceedings of the 30th annual meeting of the cognitive science society* (pp. 1041–1046). Cognitive Science Society.
45. Elsner, M., Austerweil, J. L., & Charniak, E. (2007). A unified local and global model for discourse coherence. *Proceedings of the Human Language Technologies-North American Association of Computational Linguistics (NAACL)*.
46. Charniak, E., Johnson, M., Elsner, M., Austerweil, J. L., Ellis, D., Haxton, I., Hill, C., Iyengar, S., Moore, J., Pozar, M., & Vu, T. (2006). A unified local and global model for discourse coherence. *Proceedings of the Human Language Technologies-North American Association of Computational Linguistics (NAACL)*.

Peer-Reviewed Book Chapters

1. Austerweil, J. L., Pilgrim, C., & Cao, K. (2026). Burstier events: Analysing human memory over a century of events using The New York Times. In T. T. Hills & G. Pogrebna (Eds.), *The Cambridge Handbook of Behavioral Data Science* (pp. 675–705). Cambridge University Press.
2. Austerweil, J. L., Sanborn, A. N., Lucas, C., & Griffiths, T. L. (2025). Capturing the growth of knowledge with nonparametric Bayesian models. In T. L. Griffiths, N. Chater, & J. B. Tenenbaum (Eds.), *Bayesian Models of Cognition: Reverse Engineering the Mind*. MIT Press.
3. Malle, B. F., Scheutz, M., & Austerweil, J. L. (2017). Networks of social and moral norms in human and robot agents. In M. I. A. Ferreira, J. S. Sequeira, M. O. Tokhi, E. E. Kadar, & G. S. Virk (Eds.), *A world with robots: International conference on robot ethics: Icre 2015* (pp. 3–17). Springer International Publishing.
4. Austerweil, J. L., Gershman, S. J., Tenenbaum, J. B., & Griffiths, T. L. (2015). Structure and flexibility in Bayesian models of cognition. In J. R. Busemeyer, Z. Wang, J. T. Townsend, & A. Eidels (Eds.), *Oxford handbook of computational and mathematical psychology* (pp. 187–208). Oxford University Press.
5. Austerweil, J. L., & Griffiths, T. L. (2012). Human feature learning. In *Encyclopedia of the sciences of learning*. Springer.

Peer-Reviewed Manuscripts Under Review

1. Austerweil, J. L., Szep, G., Shinada, M., Zhong, Z., & Razik, F. (2026). *Who is checking and correcting? The division of oversight labor between students and agentic AI* [Manuscript under review. IRB-approved semester study of 42 students working with an agentic coding tool in two university courses (586 sessions, 10,811 exchanges, Japanese and English)].

Presentations

Invited Talks

Rule-ambiguity and distributional play in asynchronous L2 shiritori (with J. Nishikawa and J. Morita)	International Session, 43rd Annual Meeting of the Japanese Cognitive Science Society	08/31/26
Panel on AI, Health, and Disability	<i>Pro-Justice: Cross-Cultural Approaches to AI Ethics</i> Workshop, University of Tokyo (co-organized with the University of Cambridge)	03/11/26
Knowledge representation and retrieval over a lifetime	Informatics Seminar, Shizuoka University	01/09/26
Understanding knowledge using Markov Chain Monte Carlo (MCMC) with People	Computational Summer School on Modeling Social and Collective Behavior (COSMOS) at RIKEN	10/02/25
Understanding Disease Progression using Bayesian Event-Based Models	National Center of Neurology and Psychiatry (Japan)	09/18/25
Creating Something Different: Similarity, Contrast, and Representativeness	UW-Madison: Cognitive Brown Bag	11/15/2023
Understanding semantic memory retrieval as foraging human knowledge	UIUC Brown Bag	10/20/2023
Psychology of Creativity: Bilingualism and Creativity	Creativity Unleashed Cross Labs Workshop 2023 (Kyoto, Japan)	04/24/2023
Creating Something Different: Similarity, Contrast, and Representativeness	Sungkyunkwan University invited lecture	03/09/2023
False Memories and Pattern Completion in Alzheimer's Disease: A Grocery Fluency Task	2023 Research in the Rotunda	03/08/2023
Human-Machine Interactions	Kyoto University Institute for the Future of Humanity	12/13/2022
Human and Machine Learning	RIKEN AIP	10/05/2022
Teaming with state interventions: Ms. Pacman with Learning Roombas	STRONG Speculative Human-Machine Teaming Workshop	07/25/2022
Network Estimation from Fluency Tasks	VABHS Advanced Statistics Seminar	03/21/2022
Semantic Network and Fluency Utility	Goldberg Lab at Princeton University	12/13/21
The Role of Psycholinguistic Factors in Large-Scale Question Answering Systems	American Family Insurance Lecture Series	12/07/21
Knowledge representation and Retrieval	Keynote for 2021 Meeting of the Society for Mathematical Psychology	07/09/21
Is Reinforcement Just a Value to be Maximized?	Reinforcement Learning for Education	06/29/21
Knowledge Representation and Retrieval	American Family Insurance Visiting Professor Series	10/26/20
Knowledge Representation and Retrieval	AI @ UW-Madison	10/14/20
Bayesian Collaborative Cognition Approach	DARPA Social Sim PI Meeting	06/26/19
Knowledge Representation	American Family Insurance Machine Learning Group	06/14/19
Do people care about others as if they are reinforcement learners?	Computer Science Guest Lecture at Carleton College	05/31/19
Doing vs. showing. Teaching via Demonstration	Computer Science Colloquium Series at Carleton College	05/30/19
Why should AI care about people? The case of non-verbal teaching as communication	AI @ UW-Madison	04/24/19
Bayesian Collaborative Cognition	DARPA Social Simulation Program December Challenge	12/11/18
Analyzing healthy and pathological memory retrieval as (censored) random walks over different knowledge structures	UW-Madison's Computation and Informatics in Biology and Medicine Colloquium	10/09/18
Human Perception of Patterns and Randomness as Algorithmic Complexity Over Probabilistic Stack Automata	Conference on Complex Systems	09/25/18
Revealing Knowledge Impairment Due to Alzheimer's Disease Using Novel Network Science Methods	Cognitive Network Science	06/12/18

Relations Between Category Learning and Generation	Midwest Cognitive Science Conference	05/13/18
Nonverbal teaching as communication	Purdue University, Department of Psychology Colloquium	01/12/18
Nonverbal Teaching as Communication	Indiana University, Department of Psychological and Brain Sciences. Cognition Seminar Series.	10/25/17
Language Biases Our Perception of a Noisy World... And It Should	Keynote at Conference on Language and Perception	09/07/17
Uncovering unsupervised categorization biases using Markov chain Monte Carlo with People	50th Annual Meeting of the Society for Mathematical Psychology	07/23/17
Analyzing Semantic Memory Retrieval Using Network Theory	Cognitive Network Science	06/19/17
Randomly walking over human knowledge	HAMLET talk series at UW-Madison	03/11/16
Constructing symbols from raw sensory input: A computational approach	Psychology Colloquium at University of Wisconsin-Madison	03/10/16
Randomly walking over human knowledge	Cognitive and Brain Sciences Lunch Talk Series at SUNY Binghamton	02/26/16
Constructing representations using Bayesian non-parametrics and connections between human knowledge, optimal foraging, and random walks on graph	Brown University Division of Applied Mathematics' Pattern Theory Learning Lunch Seminar Series	02/03/16
Constructing symbols from raw sensory input: A computational approach	Michigan State University Psychology Department Colloquium	01/20/16
Constructing symbols from raw sensory input: A computational approach	University of Michigan Psychology Department Colloquium	12/19/15
Constructing symbols from raw sensory input: A computational approach	NYU Cognition and Perception Seminar	11/12/15
Constructing symbols from raw sensory input: A computational approach	Harvard University's Cognition, Brain, and Behavior's Brown Bag Series	10/29/15
An Overview of Representation Construction in People and Computers	Eastern Psychological Association	03/15/14
Constructing context-sensitive representations: A case study with features	Workshop for Implications of Bayesian Cognitive Models for the Intelligence Community	12/12/13
Understanding how people construct flexible representations: A computational approach	Brown Institute for Brain Science	11/14/13
Understanding how people construct flexible representations: A computational approach	Brown University	01/22/13
Constructing flexible representations	University of Colorado, Boulder	12/14/12
Constructing flexible feature representations using nonparametric Bayesian inference	The Uncertainty in Natural Intelligence Workshop at Uncertainty in Artificial Intelligence	08/18/12
Why your mom was wrong about playing with your food: Bayesian nonparametric methods	Invited talk in the Bayesian Methodology symposium at The 45th Annual Meeting of the Society for Mathematical Psychology	07/22/12
Constructing flexible feature representations using nonparametric Bayesian inference	University of Edinburgh	03/23/12
Constructing flexible feature representations using nonparametric Bayesian inference	University of California, Berkeley	03/01/12
Effects of context on object representation: a computational exploration	Stanford University	05/19/11
Effects of context on object representation: a computational exploration	New York University Cognition Area	11/23/10
Effects of context on object representation: a computational exploration.	University of California, Berkeley	09/14/10

The effect of distributional information on feature learning	Stanford University	04/17/09
How the human perceptual system learns (some) variables	McDonnell Workshop on Problems of Variable Definition and Selection at CMU	11/05/08

Abstract-Based Conference Presentations

Hao, H., & Austerweil, J. L.	Bayesian Event-Based Model for Disease Subtype and Stage Inference	Australasian Mathematical Psychology Conference (AMPC)	02/20/26
Payton, M., Vitello, V., & Austerweil, J. L.	Rethinking Rumination: A Decision-Theoretic Approach Without Negativity Bias	Annual Meeting of the Cognitive Science Society	08/01/25
Hao, H., Prabhakaran, V., Nair, V. A., Adluru, N., & Austerweil, J. L.	Stage-Aware Event-Based Modeling (SA-EBM) for Disease Progression	3rd Bayes Duality Workshop at RIKEN AIP	06/25/25
Austerweil, J. L., & Cao, K. C.	Teaching Agents with Physical Intervention	2023 Australasian Society for Mathematical Psychology	02/09/23
Austerweil, J. L., Cao, K., Bockhurst, J., & Fung, G.	The Role of Psycholinguistic Factors in Large-Scale Question Answering Systems	62nd Annual Meeting of the Psychonomic Society	07/06/21
Chuang Y.-S., Hubbard, E. M., & Austerweil, J. L.	The "Fraction Sense" Emerges from a Deep Convolutional Neural Network	42nd Annual Meeting of the Cognitive Science Society	07/30/20
Rao, A. N., & Austerweil, J. L.	Interactions Between Categorization and Intuitive Physics	42nd Annual Meeting of the Cognitive Science Society	07/30/20
Cao, K., Afrasiabi, M., & Austerweil, J. L.	How does modern life affect memory retrieval: Analyzing news headlines	53rd Annual Meeting of the Society for Mathematical Psychology	07/27/20
Sanborn, A. N., Heller, K., Austerweil, J. L., & Chater, N.	REFRESH: A new approach to modeling dimensional biases in perceptual similarity and categorization	53rd Annual Meeting of the Society for Mathematical Psychology	07/27/20
Liew, S. X., & Austerweil, J. L.	Distinguishing new from not-old categories	Australian Mathematical Psychology Conference	02/16/19
Austerweil, J. L., Griffiths, T. L., Daniels, D., & Tenenbaum, J. B.	Human Perception of Patterns and Randomness As Algorithmic Complexity Over Probabilistic Stack Automata	Conference on Complex Systems	09/15/18
Malle, B. F. & Austerweil, J. L.	From value alignment to norm competence: How robots and AI will conform to human society	APA's Conference on Technology, Mind, and Society	04/06/18
Zemla, J. C., & Austerweil, J. L.	Modeling Semantic Fluency Data in Alzheimer's Patients	2018 Alzheimer's Disease and Related Disorders Research Day, Madison, WI	03/23/18
Austerweil, J. L. & Zemla, J. C.	Estimating an Individual's Network Efficiently from Human Fluency Data	58th Annual Meeting of the Psychonomic Society	11/09/17
Zemla, J. C. & Austerweil, J. L.	Modeling Semantic Fluency Data in Alzheimer's Patients	58th Annual Meeting of the Psychonomic Society	11/09/17
Miller, H., Jensen, C., Austerweil, J. L., & Simmering, V.	Investigating flexibility in young children's spatial recall strategies through probabilistic modeling	2017 Meeting of the Cognitive Development Society	10/14/17
Ho, M. K., Littman, M., MacGlashan, J., Cushman, F., & Austerweil, J. L.	Showing versus Doing: Teaching by Demonstration	Inaugural Conference on Computational Cognitive Neuroscience	09/07/17

Austerweil, J. L.	Uncovering unsupervised categorization biases using Markov chain Monte Carlo with People	50th Annual Meeting of the Society for Mathematical Psychology	07/24/17
Zemla, J. C., & Austerweil, J. L.	Estimating semantic networks from fluency data	50th Annual Meeting of the Society for Mathematical Psychology	07/24/17
Kleiman-Weiner, M., Ho, M. K., Austerweil, J. L., Littman, M., & Tenenbaum, J. L.	Learning to Cooperate and Compete	The Multidisciplinary Conference on Reinforcement Learning and Decision Making	06/12/17
Sobel, D. M., & Austerweil, J. L.	Coding choices affect the analysis of a false belief measure	2017 Meeting of the Society for Research in Child Development	04/07/17
Ho, M. K., Littman, M. L., MacGlashan J., Cushman F., & Austerweil J. L.	Human Teaching by Demonstration: Showing versus Doing Reinforcement Learning Tasks.	11th Annual Machine Learning Symposium of the New York Academy of Sciences	03/03/17
Kenett, Y. M., Allaham, M. M., Austerweil, J. L., & Malle, B. F.	The Norm Fluency Task: Unveiling the Properties of Norm Representation	57th Annual Meeting of the Psychonomic Society	11/19/16
Montambault, B., Lucas, C., & Austerweil, J. L.	The construction of function representations	38th Annual Conference of the Cognitive Science Society	08/12/16
Conaway, N., Austerweil, J. L., & Kurtz, K.	Two Categories for the Price of One: Generating a Contrast Category after Single-Category Learning.	49th Annual Meeting of the Society for Mathematical Psychology	08/08/16
Ho, M. K., Littman, M. L., Cushman, F., & Austerweil, J. L.	Not Quite Intuitive Behaviorists: Teachers Use Rewards and Punishments Communicatively and Not as Reinforcement.	49th Annual Meeting of the Society for Mathematical Psychology	08/05/16
Ho, M. K., Austerweil, J. L., Littman, M., & Cushman, F.	Generous Teachers: Pedagogical Rewards as Reinforcement Versus Communication	17th Annual Convention of SPSP	01/30/16
Austerweil, J. L., & Qian, T.	Additive or substitutive? Constructing features, their types and values from observations.	56th Annual Meeting of the Psychonomic Society	11/20/15
Austerweil, J. L.	Dissociating Top-down and Bottom-up Theories of Autism Spectrum Disorders Using Bayesian Models	Rhode Island Consortium for Autism Research & Treatment	09/07/15
Xie, W. & Austerweil, J. L.	The role of relations and attributes in inductive inference	48th Meeting of the Society for Mathematical Psychology	07/19/15
Austerweil, J. L., & Qian, T.	Fast and easy to use inference method for Bayesian nonparametric processes	48th Meeting of the Society for Mathematical Psychology	07/19/15
Ho, M. K., Littman, M. L., Cushman, F., & Austerweil, J. L.	Teaching Behavior with Punishments and Rewards	The 2nd Meeting of the Multidisciplinary Conference on Reinforcement Learning and Decision Making	06/08/15
Austerweil, J. L., Stanworth, K., & Franklin, A.	Dissociating Top-down and Bottom-up Theories of Autism Spectrum Disorders Using Bayesian Models	The 55th Annual Meeting of the Psychonomic Society	11/21/14
Austerweil, J. L., Stanworth, K., & Franklin, A.	Dissociating Top-down and Bottom-up Theories of Autism Spectrum Disorders Using Bayesian Models	47th Meeting of the Society for Mathematical Psychology	06/20/14
Austerweil, J. L., & Griffiths, T. L.	Learning how to generalize	The 53rd Annual Meeting of the Psychonomic Society	11/17/12

Prinzmetal, W., Whiteford, K., Austerweil, J. L., & Landau, A. N.	When information matters: The effects of cue predictability and distractors on the allocation of attention	The 12th VSS Annual Meeting	05/12/12
Austerweil, J. L., Griffiths, T. L., & Palmer, S. E.	Learning the orientation dependence of ambiguous images through feedback	Configural Processing Consortium	11/2/11
Austerweil, J. L., & Griffiths, T. L.	A computational framework for inferring feature representations	52nd Annual Meeting of the Psychonomic Society	11/3/11
Ying, J., Austerweil, J. L., & Griffiths, T. L.	Why every pop song sounds the same: Revealing melodic expectations through cultural transmission	44th Annual Meeting of the Society for Mathematical Psychology	06/18/11
Austerweil, J. L. & Griffiths, T. L.	Feature learning as nonparametric Bayesian inference	43rd Annual Meeting of the Society for Mathematical Psychology	08/08/10
Austerweil, J. L., & Griffiths, T. L.	Feature Learning Using Statistics	16th Meeting of the Cognitive Science Association for Interdisciplinary Learning	07/26/10
Austerweil, J. L. & Griffiths, T. L.	Understanding how people learn the features of objects as Bayesian inference	10th VSS Annual Meeting	05/11/10
Austerweil, J. L. & Griffiths, T. L.	Analyzing human feature learning as non-parametric Bayesian inference	14th Meeting of the Cognitive Science Association for Interdisciplinary Learning	08/03/08

Advising

Highlights:

- * Mentored advisees continue into leadership positions in academia or industry.
- * Two postdoctoral advisees are starting tenure-track faculty positions in Fall 2020 (Syracuse University and Technion).
- * One former PhD student who received his PhD and our research is published in *Journal of Experimental Psychology: General*, and received a talk at *Neural Information Processing Systems* (<2% of papers).
- * Three published journal articles with undergraduate co-authors.
- * Co-trainer for K99 awarded to Jet Vonk.

Scientists, Researchers, and other Postdoctoral Scholars

Zhuolun Zhong	2025-	Research Scientist, Computational Mind Laboratory, Chiba Tech; begins PhD April 2027
Mohsen Afrasibi	Sep 18 - Dec 20	
Jeffrey Zemla	Oct 16 - Aug 20	Assistant Professor at Syracuse University
Nicole Beckage	Jan 18 - Sep 18	Intel
Shi Xian Liew	Jan 18- Oct 19	Postdoctoral Researcher with Ben Newell at University of New South Wales
Nolan Conaway	Aug 16 - Jul 17	Shutterstock
Yoed Kenett	Oct 15 - Jul 16	Assistant Professor at Technion
Ting Qian	Jul 14 - Apr 15	Lecturer/Senior Lecturer in Data Science at Princeton University

Graduate Students

Michael Payton	PhD Aug 2026	Dissertation: <i>A Computational Framework for Perseverative Thought</i>
Hongtao Hao	PhD May 2026	Disease progression modeling (TEMPO, SA-EBM); https://hongtaoh.com
Mark Ho	PhD Dec 2017	Assistant Professor of Psychology, New York University
Kesong Cao	Masters (Psychology) Spring 2025	

Post-Baccalaureate and Undergraduates

Long-term Research Assistants

Samantha Greco (Undergraduate RA; Presenting at UW Research in the Rotunda 2023; L&S Honors Grant; Hillsdale Research Fellowship)

Blake Chambers (Lab Manager Leading Grocery Shopping Tasks for Studying Healthy and Unhealthy Aging at a Microlongitudinal Scale)

Yiwen Weng (Analysis Support for Project Analyzing L2 Learning using Duolingo Data with Elise Hopman; Now Harvard University Masters in Data Science)

Ananthao Rao (L&S Senior Thesis Grant; Hillsdale Undergraduate/Faculty Research Fellowship; Bromley Travel Grant)

Kesong Cao (L & S Senior Thesis Grant; Welton Summer Sophomore Apprenticeship; Co-author on 2020 *BRM* paper)

Sam Westby (L & S Senior Thesis Grant; Now PhD Student at Northeastern University)

Kendra Lange (L&S Senior Thesis Grant; First author peer-reviewed publication; Now PhD Student at Northeastern University)

Jacqueliene Erens (Hillsdale Undergraduate/Faculty Research Fellowship; Now PhD Student at UIUC)

Elizabeth Pettit (Now PhD Student at Miami University)

Carol Medina (Now PhD Student at University of Chicago)

Dylan Daniels (co-author on 2018 *Cognitive Psychology* paper)

Emily Gleason (Summer Research Award; Dean of the College's Prize in Cognitive Science)

Shorter-term Research Assistants: Zafieri Zaniel, Cali Stanoch, Saron Setotaw, Xinhui Yu, Hengijiali Xu, Atulya Reddy, Shri Shruti Shridhar, Dong Fang, Wenquan Chen, Chongyue Zhao, Jonathan Brandenburg, Qiuxuan Wu, Sichao Wang, Willa Cai, Yilin Liu, Quixuan Wu, Phoenix Kim, Sophie Sandweiss, Rebekah Manweiler, Vincent Kubala, Emily Low, Wenting Xie, Christine Whalen, Philip Bold, Tariq Abu-Akeel, and Julie Helmers

Service

Department/University

Undergraduate Research Scholars Mentor	Fall 2020-
Chair of Workgroup on BIPOC Community Building	Fall 2020
Department Faculty Senator	Fall 2020-Spring 2021
University Student Misconduct Committee	Fall 2018-
Lecture and Ongoing Consulting for the Office of Student Financial Aid	Summer 2018-Winter 2020
Madison eSports Faculty Advisor	Fall 2017
Alternate Senator to faculty Senate	Fall 2017-Spring 2018
Department Graduate Committee	Fall 2016-Spring 2018
Social Psychology Job Search Committee	Fall 2014-Spring 2015
Graduate Program Restructuring Committee	Fall 2014-Spring 2015
Cognitive Science Undergraduate Self-Study Committee	Spring 2014
Psychology Undergraduate Honors Award Committee	Spring 2014

Student Committees

Harrison Rosenberg	EECS Committee	Fall 2019 -
Yun-Shiuan Chuang	FYP Co-Advisor	Fall 2019 -
Kushin Mukherjee	FYP Committee	Fall 2019 -
Yugang Duan	FYP Committee	Fall 2019 -
Qiawen Liu	FYP Committee	Fall 2019 -
Michael Payton	FYP and Prelim Advisor	Spring 2019 -
Ian Johnston	Prelim Committees	Fall 2018 -
Sarah Sant'Ana	FYP and Prelim Committees	Fall 2018 -
Vincent Frigo	FYP Committee	Summer 2019 -
Sounak Mohanta	FYP Committee	Fall 2018-Fall 2019
Quan Wan	FYP Committee	Fall 2018-Fall 2019
Ron Pomper	Prelim Committee	Fall 2018-Spring 2019
Mark Koranda	Prelim and Dissertation Committee	Fall 2017 -
Elise Hopman	Prelims and Dissertation Co-Advisor	Spring 2017 -

Steve Schwering	FYP and Prelim Committees	Spring 2017 -
Martin Zettersen	Prelim and Dissertation Committees	Spring 2017 - Spring 2020
Mark Ho	Main PhD Advisor	Mar 2014-Dec 2017
Pablo Caceres	FYP and Breadth Prelim Committees	Spring 2017 - Spring 2020
Olivia Zhao	FYP Committee	Spring 2017 - Fall 2018
Adrienne Wood	Dissertation Committee	Fall 2016 - Spring 2017
Elena Luchkina	Prelim Committee	Fall 2015-Spring 2016
Boyoung Kim	Prelim and Dissertation Committee	Fall 2015-Spring 2016
Alexander Fengler	First-Year Advisor	Fall 2015-Spring 2016
Babak Hemmatian	First-Year Committee	Fall 2015-Spring 2016
Junkyung Kim	First-Year Committee	Fall 2014 - Spring 2016
Matthew Ricci	First-Year Committee	Fall 2014- Spring 2016
Xi Yang	Masters' Committee	Fall 2014- Spring 2015
Ceyda Sayali	First-Year Committee	Fall 2014- Spring 2015
Gideon Goldin	Dissertation Committee	Fall 2013

Community

NAMI Public Showing of "The S Word"	02/29/20
Lecture at the Monona Senior Center	02/07/20
Ongoing Research Collaboration with American Family Insurance	Summer 2019-
Public Panel on Misinformation	Summer 2019
Public Lecture at Wisconsin Institute for Discovery (eLUCID8)	08/07/18
Volunteer Advising for Veda Data Solutions	Winter 2018 - Fall 2019
Wisconsin Idea Seminar	Summer 2017

Professional

Editor	Associate Editor for <i>Computational Brain & Behavior</i>	Summer 2019-
RISC	Faculty Mentor for DoD and IC Summer Internship	Summer 2023
Conference Organizing	Co-Organizer for the 2018 Annual Meeting of the Society for Mathematical Psychology	Winter 2017-Summer 2018
Workshop and Symposium Organizing	Co-Organizer for <i>Make It So (MISO) Camp</i> , Tokyo — a two-day working session on directing AI for practitioners outside software engineering, with Noah Rafor and Igor Schwarzmann, in partnership with Chiba Tech and the Henkaku Center (https://misocamp.com); 2016 NeurIPS Workshop on the Future of Interactive Learning Machines; 2014 Symposium for the Eastern Psychological Association; 2012 UAI Workshop on Uncertainty in Natural Intelligence; Co-Organizer for a 2011 Cognitive Science Symposium on Computational Constructivism	Aug 2026; Dec 2016; March 2014; Aug 2012; Jul 2011
Grant Reviewing	Ad Hoc Reviewer for ISF, NSF, and Marie Curie Foundation; NSF GRFP Panelist (multiple years)	2016-
Journal Reviewing (each journal mentioned once)	<i>Cognition</i> , <i>Cognitive Processing</i> , <i>Cognitive Psychology</i> , <i>Cognitive Science</i> , <i>eLIFE</i> , <i>IEEE: Bioinformatics</i> , <i>International Journal for Approximate Reasoning</i> , <i>Journal of the Royal Society: Interface</i> , <i>Journal of Mathematical Psychology</i> , <i>JEP: General</i> , <i>JEP:LMC</i> , <i>Nature: Human Behaviour</i> , <i>Nature: Scientific Reports</i> , <i>Open Mind</i> , <i>PLOS One: Computational Biology</i> , <i>Psychological Science</i> , <i>Psychological Review</i> , <i>Psychonomic Bulletin and Review</i> , <i>Topics in Cognitive Science</i>	2010-
Conference Paper Reviewing	<i>Artificial Intelligence and Statistics</i> , <i>Cognitive Science Society</i> , <i>IEEE:ICDL-EPIROB</i> , <i>Neural Information Processing Systems</i>	2010-
Other Conference Leadership	Cognitive Science Society Awards Committee, Cognitive Science Society Publication-based Talk Committee, Cognitive Science Society Program Committee, Cognitive Science Mentoring Program Abstract Submissions for the Society for Mathematical Psychology	2018, 2020, 2020, & 2020
Memberships:	Cognitive Science Society, Society for Mathematical Psychology, Fellow of the Psychonomic Society, and Associate of the Brain and Behavioral Sciences Journal	2008-, 2010-, 2017-, and 2017-

Teaching**Honors**

UW-Madison Distinguished Honors Faculty Award	Spring 2019
EXCEL Initiative	Fall 2018-Summer 2020
Final project from PSYC525 added to <i>Locally Sourced</i> , internal successful writing assignment handbook	Summer 2018
University Housing Honored Instructor	Spring 2018
UW-Madison Teaching and Learning Excellence Faculty Fellow	Fall 2016-Spring 2017

Regular courses

The second column lists average course and instructor effectiveness from student feedback (1 to 5 Likert scale). Note: prior to Spring 2017, lower was better (from when I was at Brown University). Ratings are not reported for Chiba Tech courses.

Antidisciplinary Project Studio I (APS-I), School of Design & Science, Chiba Tech (https://aps-i.chibatech.dev)		Spring 2026
Human and Machine Learning, School of Design & Science, Chiba Tech (https://hml.chibatech.dev)		Spring 2026
Co-Instructor, COSMOS Summer School on Modeling Social and Collective Behavior, RIKEN (https://cosmossummerschool.github.io)		2025
PSYC/COMPSCI/BMI841: Computational Cognitive Science	4.1	Fall 2019
PSYC525: Cognition in Health and Society	4.3	Spring 2019
PSYC/COMPSCI/BMI841: Computational Cognitive Science	4.7	Fall 2018
PSYC525: Cognition in Health and Society	4.22	Spring 2018
PSYC/COMPSCI/BMI841: Computational Cognitive Science	4.5	Fall 2017
PSYC501: Cognition in Health and Society	4.1	Spring 2017
CLPS0200: Human Cognition	1.5	Spring 2016
CLPS2200: Core Topics in Cognition	1.7	Spring 2016
CLPS1211: Human and Machine Learning	1.4	Fall 2015
CLPS0200: Human Cognition	1.5	Spring 2015
CLPS1200: Thinking	1.5	Fall 2014
CLPS1211: Human and Machine Learning	1.3	Spring 2014
CLPS1200: Thinking	1.4	Fall 2013

Open Educational Resources

A Narrative Introduction to Probability (<https://josephausterweil.github.io/probintro/>) — an open, in-progress tutorial that builds from set-based probability to working Bayesian nonparametric models, with six interactive Jupyter notebooks and probabilistic programming in GenJAX throughout. Maintained in English and Japanese; source at <https://github.com/josephausterweil/probintro>. Supported by the Japanese Probabilistic Computing Consortium Association (JPCCA).

Independent Studies

4 Undergraduate Independent Studies (4 Psych), and 1 Undergraduate Honors Thesis (COMPSCI)	Spring 2020
3 Undergraduate Independent Studies (3 Psych), and 1 Masters' Research (COMPSCI)	Fall 2019
1 Undergraduate Honors Thesis (Psych), and 3 Undergraduate Independent Studies (1 Psych & 2 COMPSCI)	Spring 2019
1 Undergraduate Honors Thesis (Psych), and 3 Undergraduate Independent Studies (1 Psych & 2 COMPSCI)	Fall 2018
1 Undergraduate Honors Thesis (Psych), 5 Undergraduate Independent Studies (2 Psych & 3 COMPSCI), and 1 Masters Research Study (COMPSCI)	Spring 2018
8 Undergraduate Independent Studies (2 PSYCH & 6 COMPSCI)	Fall 2017
3 Undergraduate Independent Studies (3 PSYCH)	Spring 2017
4 Undergraduate Direct Research (4 PSYCH)	Fall 2015
3 Undergraduate Direct Research (3 PSYCH)	Spring 2015
3 Undergraduate Direct Research (3 PSYCH)	Fall 2014
1 Undergraduate Direct Research (1 PSYCH)	Spring 2014